BILL OF MATERIALS (BOM) — VERIFIED

Adaptive Power Distribution System

Verified against the supplied final schematic screenshots

1. Verification Basis

The component reference designators and values below were checked directly against the three supplied schematic screenshots: the Protected 12 V section, the +5 V rail, and the +3.3 V rail. This verification corrects the previous BOM where the values of R2 and R3 were not matched to the final schematic.

2. Verified BOM

No.	Ref.	Value / Rating	Component	Qty.	Package	Function
1	J1	12V_DC_Input	2-pin connector	1	THT	12 V input
2	F1	3 A	Fuse	1	THT	Main input protection
3	D1	1N5822	Schottky diode	1	THT	Reverse-polarity protection
4	C1	100 nF	Capacitor	1	THT	Protected 12 V bypass
5	C2	470 µF	Electrolytic capacitor	1	THT	Protected 12 V bulk capacitor
6	R1	2.2 kΩ	Resistor	1	THT axial	12 V indicator current limiting
7	D2	12V_Indicator	LED	1	THT LED	12 V status indication
8	U1	LM2596S-5	Buck regulator	1	TO-263 / D2PAK	5 V regulator
9	L1	68 µH	Power inductor	1	THT	5 V buck stage
10	D3	1N5822	Schottky diode	1	THT	5 V buck diode
11	C3	330 µF	Electrolytic capacitor	1	THT	5 V output capacitor
12	C5	100 nF	Capacitor	1	THT	5 V bypass
13	R2	2.2 kΩ	Resistor	1	THT axial	5 V indicator current limiting
14	D5	5V_Indicator	LED	1	THT LED	5 V status indication
15	R5	10 Ω	Power resistor	1	High-power THT	5 V / 0.5 A load model

16	F3	0.5 A Polyfuse	Resettable fuse	1	THT	5 V / 0.5 A output protection
17	R6	5 Ω	Power resistor	1	High-power THT	5 V / 1 A load model
18	F4	1.0 A Polyfuse	Resettable fuse	1	THT	5 V / 1 A output protection
19	J4	5V_OUT_A–0.5 A	2-pin connector	1	THT	5 V / 0.5 A output
20	J3	5V_OUT_C–1.0 A	2-pin connector	1	THT	5 V / 1 A output
21	U2	LM2596S-3.3	Buck regulator	1	TO-263 / D2PAK	3.3 V regulator
22	L2	68 µH	Power inductor	1	THT	3.3 V buck stage
23	D4	1N5822	Schottky diode	1	THT	3.3 V buck diode
24	C4	330 µF	Electrolytic capacitor	1	THT	3.3 V output capacitor
25	C6	100 nF	Capacitor	1	THT	3.3 V bypass
26	R3	1 kΩ	Resistor	1	THT axial	3.3 V indicator current limiting
27	D6	3V3_Indicator	LED	1	THT LED	3.3 V status indication
28	R4	11 Ω	Power resistor	1	High-power THT	3.3 V / 0.3 A load model
29	F2	0.3 A Polyfuse	Resettable fuse	1	THT	3.3 V / 0.3 A output protection
30	J2	3V3_OUT_B–0.3 A	2-pin connector	1	THT	3.3 V / 0.3 A output

3. Critical Designator / Value Verification

Reference	Verified value	Where verified
R1	2.2 kΩ	Protected 12 V section
R2	2.2 kΩ	5 V rail
R3	1 kΩ	3.3 V rail
R4	11 Ω	3.3 V load
R5	10 Ω	5 V / 0.5 A load
R6	5 Ω	5 V / 1 A load
C1	100 nF	Protected 12 V section
C2	470 µF	Protected 12 V section

C3	330 μF	5 V rail
C4	330 μF	3.3 V rail
C5	100 nF	5 V rail
C6	100 nF	3.3 V rail
D1	1N5822	Input protection
D3	1N5822	5 V buck stage
D4	1N5822	3.3 V buck stage
D2	12V_Indicator	12 V status LED
D5	5V_Indicator	5 V status LED
D6	3V3_Indicator	3.3 V status LED
F1	3 A	Input fuse
F2	Polyfuse, 0.3 A branch	3.3 V output
F3	Polyfuse, 0.5 A branch	5 V / 0.5 A output
F4	Polyfuse, 1.0 A branch	5 V / 1.0 A output

4. Output Protection Mapping

Connector	Rail	Load	Protection
J2	3.3 V	0.3 A	F2 — 0.3 A polyfuse
J4	5 V	0.5 A	F3 — 0.5 A polyfuse
J3	5 V	1.0 A	F4 — 1.0 A polyfuse

5. Component Count Summary

Category	Quantity
LM2596 regulators	2
68 μH inductors	2
1N5822 diodes	3
Electrolytic capacitors	3 (470 μF ×1, 330 μF ×2)
100 nF capacitors	3
LED indicators	3
LED resistors	3 (2.2 kΩ ×2, 1 kΩ ×1)
Load resistors	3 (10 Ω, 11 Ω, 5 Ω)

Fuses / polyfuses	4 (3 A ×1, 0.3 A ×1, 0.5 A ×1, 1 A ×1)
Connectors	4 (J1–J4)

6. Notes Before Final Submission

- R2 is 2.2 kΩ, not 1 kΩ. R3 is 1 kΩ. These values now match the supplied final schematic.
- The three 100 nF capacitors are C1, C5 and C6.
- The two 330 μF capacitors are C3 and C4; the 470 μF input capacitor is C2.
- The three 1N5822 diodes are D1, D3 and D4. D2, D5 and D6 are the LED indicators.
- The polyfuse mapping is F2 = 3.3 V/0.3 A, F3 = 5 V/0.5 A and F4 = 5 V/1 A.
- Exact manufacturer MPNs for generic resistors, capacitors, connectors and polyfuses should only be added after the actual procurement parts are selected.

7. Regulator Package Reference

The LM2596S-5.0 and LM2596S-3.3 variants are available in TI's 5-pin DDPAK/TO-263 package. The exact footprint used on the PCB should therefore match the purchased regulator package. TI's current documentation lists LM2596S-3.3/NOPB and LM2596S-5.0/NOPB as 5-pin DDPAK/TO-263 devices.

Reference: Texas Instruments LM2596 datasheet.